



V1 - Continuity, partial derivatives Diff. Calc. of functions of several variables

Let G be an open set in $\mathbb{R}^2$
 $f: G \rightarrow \mathbb{R}$ is a function. We say f is
 Continuous at $p \in G$ if

Whenever (v_n) is a sequence in G such that
 $v_n \rightarrow p$ the corresponding real sequence
 $(f(v_n))$ converges to $f(p)$

Exercise: Formulate the definition of
 $\lim_{v \rightarrow p} f(v) = l$.

Model your def based on ideas of Chapt 2.

Examples:

- (i) $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ given by
- (a) $f(x, y) = x^2 + y^2$
 - (b) $f(x, y) = x^3 - 3x^2y + 3xy^3$
 - (c) are continuous.

Well, let $v_n = (x_n, y_n)$ be a seq in $\mathbb{R}^2$
 s.t. $v_n \rightarrow v = (x, y)$

Then $x_n \rightarrow x, y_n \rightarrow y$

$$\therefore x_n^2 + y_n^2 \rightarrow x^2 + y^2 \quad (\text{Main step})$$

$$\therefore f(v_n) \rightarrow f(v)$$

In general, Any polynomial is Continuous



Formulate def. of Continuity of
 $f: G \rightarrow \mathbb{R}$ when G is an
 open set in $\mathbb{R}^3$

Remark: We shall discuss Continuity
ONLY for functions $f: G \rightarrow \mathbb{R}$
 when

- (i) G is an open set in $\mathbb{R}^2$ (or $\mathbb{R}^3$)
- (ii) G is a closed box rectangle in $\mathbb{R}^2$
 or a closed box in $\mathbb{R}^3$

Example: $f: \mathbb{R}^3 \rightarrow \mathbb{R}; f(x, y, z) = \begin{vmatrix} x & y \\ z & x \end{vmatrix}$
 is continuous.

Example: The set of all quadratics
 $x^2 + ax + b; a, b \in \mathbb{R}^2$

can be thought of as $\mathbb{R}^2$

Identify $x^2 + ax + b$ with $(a, b) \in \mathbb{R}^2$

Sketch the set of points corresp. to those
 quadratics with equal roots. Is it a
 closed set?

More examples: $f: \mathbb{R}^2 \rightarrow \mathbb{R}; f: \mathbb{R}^3 \rightarrow \mathbb{R}$
 $f(v) = \|v\|$

$$\|v\| = \sqrt{x_1^2 + x_2^2} \text{ if } v = (x_1, x_2)$$

$$\|v\| = (x_1^2 + x_2^2 + x_3^2)^{1/2} \text{ if } v = (x_1, x_2, x_3)$$

Show that in both cases f is continuous

Simple: $|\|v\| - \|w\|| \leq \|v - w\|$

(Tutorial prob) (how?)



(ii) $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$f(x,y) = \frac{x^2 y}{x^2 + y^2} \quad \text{if } (x,y) \neq (0,0)$$

$$= 0 \quad \text{if } (x,y) = (0,0)$$

Is f cont. at the origin?

At other points??

proof. $|f(x,y)| = \frac{|x|}{2} \cdot \frac{|2xy|}{x^2 + y^2}$

$$\leq \frac{|x|}{2}$$

So Continuity at $(0,0)$ follows by Sandwich Thm.

Usual Rules go through:

If G is an open set in $\mathbb{R}^2$ or $\mathbb{R}^3$

or G is a closed rectangle / box

$f, g: G \rightarrow \mathbb{R}$ are two functions s.t.

f, g are Continuous at $p \in G$

then

$f+g, fg$ are Continuous at p .

If $g \neq 0$ throughout G

then $\frac{f}{g}: G \rightarrow \mathbb{R}$ is Cont. at p .

Let us return back to the last problem.

$\frac{x^2 y}{x^2 + y^2}$ is Continuous on $\mathbb{R}^2 - (0,0)$ by the last clause.

In particular, if $v \neq 0$ and $v_n \rightarrow v$

then $v_n \in B_r(v)$ for all $n \geq n_0$

Where $r > 0$ is so small that $0 \notin B_r(v)$



All the v_n enter $B_r(v)$
after a finite stage

So $f(v_n) \rightarrow f(v)$ since
the function in question is continuous on
 $\mathbb{R}^2 - (0,0)$.

In short, to discuss continuity of f
at a point $p \in G$

You just have to enclose p in a
ball $B_r(p)$ of tiny radius r and
concentrate on what's going on inside $B_r(p)$

Example: $f(x,y) = \frac{xy}{x^2+y^2}$ if $(x,y) \neq (0,0)$
 $= 0$ if $(x,y) = (0,0)$

$\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ doesn't exist

Suppose the limit exists.

Take $v_n = (\frac{1}{n}, \frac{1}{n}) \rightarrow (0,0)$

(v_n) approaches origin ALONG a
line (the line is $y=x$)

$f(v_n) = \frac{1}{2}$ for all n

Next, let us approach origin along a
different line: $(\frac{1}{n}, \frac{2}{n}) = w_n$

$f(w_n) = \frac{2}{5}$ for all n



We do not have a unique l s.t.
 $f(v_n) \rightarrow l$ for all sequences
 $v_n \rightarrow (0,0)$.

Next Example is Even more interesting:

Example: $f(x,y) = \frac{x^2y}{x^4 + y^2} \quad (x,y) \neq (0,0)$
 $= 0 \quad \text{if } (x,y) = (0,0)$

Try taking sequences $v_n \rightarrow 0$
 along different lines say $y = kx$
 $v_n = \left(\frac{1}{n}, \frac{k}{n} \right)$

What is $f(v_n)$? $\lim_{n \rightarrow \infty} f(v_n) = ?$

Can you conclude

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0 ?$$

Is f cont. at origin? No!

Try approaching origin along a parabola
 (Tutorial Problem)

Properties of Continuous functions:

Let $f: G \rightarrow \mathbb{R}$ be continuous
 G open and $p \in G$, $f(p) > 0$

Then $\exists \epsilon > 0$ such that

Theorem A'

$$B_\epsilon(p) \subset G \text{ and}$$

$$f(x) > 0 \text{ for all } x \in B_\epsilon(p)$$

In short $f(p) \text{ pos} \Rightarrow f$ stays positive in
 a neighborhood of p .

Exercise: Formulate theorems B', C', D' that parallel theorems B, C, D of Chapter 2

For B', D' take G to be a closed bounded rectangle / Box

For Thm C' take G to be a Box or a rectangle (open or closed doesn't matter). Does Convexity help?

Exercise: Prove Thm D' using Thm B'
Is it possible to prove Thms B', C' using Method of bisection? Try!!

Let G be a closed box; $f: G \rightarrow \mathbb{R}$

$$M = \sup \{ f(v) / v \in G \}$$

$$m = \inf \{ f(v) / v \in G \}$$

$M - m$ is called the spread of f on G

You may be anticipating that Thm E,
Thm F on p. 72.5

Also generalize.

In fact Thm E Directly generalizes!

Thm F (The FTC) has a Multi variable analogue and it is called the Gauss's Divergence-Thm / Stokes-Thm which will be the last Thm in this course - and the MOST Imp. one too!

Note however: Thm B' needed! Too fundamental

Partial derivatives:

Let G be open in $\mathbb{R}^2$ (or $\mathbb{R}^3$)
 $f: G \rightarrow \mathbb{R}$ be a function
 and $p = (p_1, p_2) \in G$ be a point in G

$$\lim_{h \rightarrow 0} \frac{f(p_1+h, p_2) - f(p_1, p_2)}{h}$$

if it exists is called the first partial derivative of f at p and denoted by
 $\frac{\partial f}{\partial x_1}(p)$.

Let us look at this slightly differently.

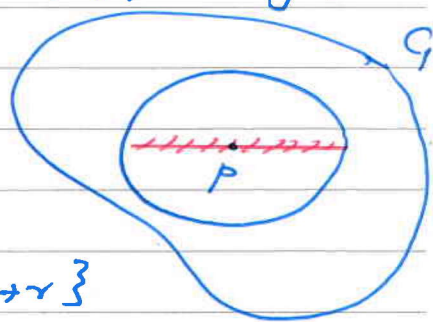
Take a disc $B_r(p) \subset G$

Work on $B_r(p)$

Consider the horizontal

slice through p :

$$\{(t, p_2) \mid p_1 - r < t < p_1 + r\}$$



Restrict the function f to this slice.

$$g(t) = f(t, p_2): \quad p_1 - r < t < p_1 + r$$

g is a function of one variable defined on the
Open interval $(p_1 - r, p_1 + r)$

We are simply computing $g'(p_1)$

In other words,

Keep the 2nd coordinate fixed at p_2
 and diff. wrt "first variable"

and this is $\frac{\partial f}{\partial x_1}(p)$

Similarly $\frac{\partial f}{\partial x_2}(p)$ is by definition

$$\lim_{h \rightarrow 0} \frac{f(p_1, p_2 + h) - f(p_1, p_2)}{h} \text{ if it exists.}$$

Example: $\sec^3 x (y + x^2)$

defined on a tiny disc $B_r(0,0)$
 $r > 0$ is so small that $|x(y + x^2)| < \pi/2$

When $(x, y) \in B_r(0,0)$. (So that f is defined)

Let us compute

$$\frac{\partial}{\partial x} \sec^3 x (y + x^2) \Big|_{(0,0)} \quad \text{and}$$

$$\frac{\partial}{\partial y} \sec^3 x (y + x^2) \Big|_{(0,0)}$$

For the first we are allowed to set $y = 0$
 obtaining $\sec^3 x^3$ and we proceed to
 diff this wrt x at $x = 0$.

$$\text{Ans} = 0. \quad (\text{Ex})$$

Next, for the second

$$\frac{\partial}{\partial y} \sec^3 x (y + x^2) \Big|_{(0,0)}$$

We first set $x = 0$ obtaining a
 function of y alone which we then
 differentiate.

Well, setting $x = 0$ gives const function 1
 whose der. wrt y is 0.



Examples from Thomas Finney (9th Ed)

Study the solved examples on pp 926-927.

p 931 Ex list 12.3 problem 7 (modified)

$f(x,y,z) = \sqrt{x^2+y^2+z^2}$ defined on $\mathbb{R}^3$
 $f(x,y,z)$

$$\frac{\partial f}{\partial x} = \frac{x}{\sqrt{x^2+y^2+z^2}} \quad \text{etc;}$$

We shall use the letter $r = \sqrt{x^2+y^2+z^2}$
 distance of x from the origin.

~~Is it~~ Do the partial der. exist at origin?

$$\frac{f(x, 0, 0) - f(0, 0, 0)}{h} = \frac{|h|}{h}$$

Clearly $\lim_{h \rightarrow 0} \frac{|h|}{h}$ doesn't exist and we conclude

$\frac{\partial f}{\partial x}$ doesn't exist at the origin.

Similarly the other two partial der. do not exist at the origin

Exercise:

Discuss whether the partial derivatives of the function $f(x,y,z) = r^3$ exist at the origin. What about $f(x,y,z) = \begin{cases} r^2 \sin \frac{1}{r} & r \neq 0 \\ 0 & \text{if } r = 0 \end{cases}$

$$r = (x^2+y^2+z^2)^{1/2}.$$

Calculate all the three partial derivatives at a general point of $f(x,y,z) = r^{-1}; \mathbb{R}^3 - \{0\}$
 and $f(x,y) = \log r; \mathbb{R}^2 - \{0\}$
 (origin removed)

Do the exercises on p 93) of
Thomas Finney

Exercise (Tutorial) $M(x,y) = \frac{y}{x^2+y^2}$
 $N(x,y) = \frac{-x}{x^2+y^2}$ on $\mathbb{R}^2 - (0,0)$

Calculate: $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}$

Homogeneous functions: A function
 $f: \mathbb{R}^3 - (0) \rightarrow \mathbb{R}$

(or $\mathbb{R}^2 - (0) \rightarrow \mathbb{R}$) is said to be homogeneous
of degree k if

$$f(tv) = t^k f(v) \text{ for all } v \in \mathbb{R}^3 - (0)$$

Thus, M, N in the last exercise are both
homogeneous of degree -1

Calculate: $x \frac{\partial M}{\partial x} + y \frac{\partial M}{\partial y}$; $x \frac{\partial N}{\partial x} + y \frac{\partial N}{\partial y}$.

$f(x,y,z) = x^3 - 3xyz + y^3 + z^3$
is homogeneous of degree 3

Compute $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z}$

Try out similar thing with $\tan^{-1}\left(\frac{x^2-y^2}{x^2+y^2}\right)$
(which is hom. deg. 0)

Can you formulate a general result?

Exercise: Formulate a general result based
on these experiments.

Prove the thm for polynomials



Example: $f(x,y) = \frac{xy}{x^2+y^2}, (x,y) \neq (0,0)$
 $= 0 \text{ if } (x,y) = (0,0)$

We know f is not Cont. at the origin
 (See p. 87)

Does $\frac{\partial f}{\partial x}$ exist at $(0,0)$?

Well, $\frac{f(h,0) - f(0,0)}{h} = \frac{f(h,0)}{h} = 0$

So yes, $\frac{\partial f}{\partial x}$ exists at the origin and has value 0

Likewise $\frac{\partial f}{\partial y}(0,0)$ exists and $= 0$.

So Existence of partial derivatives of f at $p \not\Rightarrow$ Continuity of f at p .

Exercises Problems from A.E. Taylor and
 W.R. Mann. Adv. Calc. p 199

(1) $f(x,y) = \frac{x^4+y^4}{x^2+y^2}$ if $(x,y) \neq (0,0)$; $f(0,0) = 0$

f is Cont. at origin. Partial derivatives
 $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ exist at all points.

Examine whether $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ Cont. at origin.

(2) $f(x,y) = \frac{x^3-y^3}{x^2+y^2}$ if $x^2+y^2 \neq 0$
 $f(0,0) = 0$.

Answer the same questions.

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$$(3) \quad f(x,y) = (x^2+y^2) \sin \frac{1}{\sqrt{x^2+y^2}} \\ (x,y) \neq (0,0) \\ f(0,0) = 0.$$

Show $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ exist at all points.

Show that $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ are discontinuous at $(0,0)$

Remark: One defines a notion of differentiability (we shall not do so in this course - see any book. E.g. Taylor-Mann from which this example is taken, p. 197)

This example shows that this f is differentiable at origin

Both partials discont. at origin

In Example (2) f is not diff. at origin though both partial der. exist.

both In case the partial derivatives all exist and are continuous - then the function is differentiable (as explained on p. 197 of Taylor Mann)

Thus

(a) Both partials exist (weak)

(b) Function is differentiable (Strong)

(c) Both partials exist and continuous (VERY Strong)

$$(c) \Rightarrow (b) \Rightarrow (a)$$

Now In our Course We shall NEVER have occasion to use (b) and so we do not define it.

We shall always demand (c) in all the principal theorems

Such as Green's Thm, Div Thm etc;

(End of Remark)

Higher order partial derivatives

Suppose G is an open set in $\mathbb{R}^2$ or $\mathbb{R}^3$
 $f: G \rightarrow \mathbb{R}$ is a function such that all the first order partial derivatives exist at each point of G so that we have the new functions $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ (and in case $G \subset \mathbb{R}^3$ $\frac{\partial f}{\partial z}$ also)

Question is of course

whether $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ etc, admit partial derivatives.

namely whether $\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right)$, $\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$ etc;

exist at a point $p \in G$

If so, we denote them by $\frac{\partial^2 f}{\partial x^2}(p)$,

$\frac{\partial^2 f}{\partial y \partial x}(p)$ etc;

Now it may happen that

$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)$ and $\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$ both exist at $p \in G$

but are unequal!

This is RATHER UNPLEASANT !!

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We want Conditions that would guarantee equality of

$$\frac{\partial^2}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \text{ at } p \in G.$$

We have a simple criterion which suffices for our purpose. We state it for two var. but its clear how to gen.

Thm 6.1 $f: G \rightarrow \mathbb{R}$ is a function defined on an open set containing point p . In a certain disc $B_r(p)$ around p

$\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ exist and are cont. on $B_r(p)$

$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right), \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$ are defined

on $B_r(p)$ and continuous at p .

Then,

$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)$ and $\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$ are equal at p

Remark: One can try and weaken some of the hypotheses but we are NOT

INTERESTED in fine tuning. | See AE Taylor and This version suffices.

W. R. Mann: Adv. Calc. 199 ff

In fact in all our applications we shall have $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ exist cont. on $B_r(p)$

and $\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}, \frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y \partial x}$ would

all exist and be cont. on $B_\delta(p)$.

We shall only give one example
Taylor-Mann p. 201

$$f(x,y) = \begin{cases} xy \left(\frac{x^2-y^2}{x^2+y^2} \right) & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

$$\frac{\partial f}{\partial y} \quad \frac{\partial f}{\partial x} (0,y) = -y; \quad \frac{\partial f}{\partial y} (x,0) = x$$

$$\text{Thus } \frac{\partial^2 f}{\partial y \partial x} = -1; \quad \frac{\partial^2 f}{\partial x \partial y} = 1$$

In future we shall assume always that f satisfies conditions for equality of Mixed partials to hold.

The Laplacian: Suppose $f: G \rightarrow \mathbb{R}$ is a function such that-

$$\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \text{ exist throughout } G.$$

$$\frac{\partial^2 f}{\partial x^2} \text{ and } \frac{\partial^2 f}{\partial y^2} \text{ exist throughout } G$$

$$\text{Then } \Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}. \text{ Similarly in } \mathbb{R}^3 \dots$$

Ex: Compute Δf when

$$f(x,y,z) = r^\alpha; \quad r = \sqrt{x^2+y^2+z^2} \text{ defined on } \mathbb{R}^3 - \{0\}$$

For what value(s) of α is it true that $\Delta f = 0$.

Do the problem in $\mathbb{R}^2$ also.

Compute $\Delta(\log r)$
 $r = \sqrt{x^2 + y^2}$ on $\mathbb{R}^2 - \{0\}$

Exercise:

Determine all 2nd order homogeneous polynomials $P(x, y)$ s.t. $\Delta P = 0$.

Do it for 3rd order homogeneous polynomial

Ex: Do the problems on p 931 of
 Thomas - Finney (9th Edition)

Thm 6.2: Let G be an open set in $\mathbb{R}^2$
 $u, v: G \rightarrow \mathbb{R}$ be two functions
 that are differentiable admit partial
 derivatives $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$ on G

are continuous and second order derivatives
 exist and continuous. ^{all}

Suppose $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ (*)

Then $\Delta u = 0$ and $\Delta v = 0$.

proof: Easy.

Example: (i) $u(x, y) = e^x \cos y$
 $v(x, y) = e^x \sin y$

Examine if (*) holds.

(ii) $u(x, y) = \cos x \cosh y$. Find v such
 that (*) holds.